Systems of Linear Equations

Section 1: Augmented Matrices and Row Echelon Form (REF)

1 Which matrix represents a valid Reduced Row Echelon Form (RREF)?

a. $[[1, 0, 3], [0, 1, -2], [0, 0, 0]]$

b. $[[1, 2, 3], [0, 1, -2], [0, 0, 0]]$

c. $[[2, 0, 0], [0, 1, 0], [0, 0, 1]]$

d. $[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$

2 How many pivots does a 3×5 matrix have at most?

a. 3

b. 5

c. 2

d. 15

3 An augmented matrix has a row of the form $[0, 0, \ldots, 0 \mid b]$ with $b \neq 0$. What does this establish about the linear system?

a. The system is inconsistent (has no solution)

b. The system has a unique solution

c. The system has infinitely many solutions

d. The determinant is equal to b

Section 2: Solution Sets and Free Variables

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4 A consistent system has 5 variables and 3 pivot columns in its coefficient matrix. How many free variables does the system have?

a. 2

b. 3

c. 5

d. 0

5 When does a linear system $A\mathbf{x} = \mathbf{b}$ possess a unique solution?

a. When the system is consistent and every column of A contains a pivot (no free variables)

b. Whenever the number of equations equals the number of variables

c. Whenever $\mathbf{b} \neq \mathbf{0}$

d. When there is at least one free variable

6 If a system is consistent and has at least one free variable, how many solutions exist over $\mathbb{R}$?

a. Infinitely many

b. Exactly one

c. None

d. Finite number equal to the number of free variables

Section 3: Homogeneous Systems and Null Space

7 What is always true about the homogeneous linear system $A\mathbf{x} = \mathbf{0}$?

a. It is always consistent, because $\mathbf{x} = \mathbf{0}$ is always a solution (the trivial solution)

b. It has no solutions if $\det(A) = 0$

c. It always has infinitely many solutions

d. It only has solutions when A is square

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8 Under what condition does the homogeneous system $A\mathbf{x} = \mathbf{0}$ have non-trivial solutions (solutions where $\mathbf{x} \neq \mathbf{0}$)?

a. If and only if the system has at least one free variable

b. If and only if A has more rows than columns

c. If and only if all entries of A are positive

d. Whenever $\mathbf{b} = \mathbf{0}$

9 The set of all solutions to $A\mathbf{x} = \mathbf{0}$ forms a subspace of $\mathbb{R}^n$ called the:

a. Null space (or kernel) of A

b. Column space of A

c. Row space of A

d. Span of $\mathbf{b}$

Section 4: Matrix Invertibility and Systems

10 According to the Invertible Matrix Theorem, for a square $n \times n$ matrix A, which statement is equivalent to A being invertible?

a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$

b. The columns of A are linearly dependent

c. $\det(A) = 0$

d. The null space of A contains non-zero vectors

11 If A is invertible and $A\mathbf{x} = \mathbf{b}$, how is the unique solution expressed algebraically?

a. $\mathbf{x} = A^{-1}\mathbf{b}$

b. $\mathbf{x} = \mathbf{b}A^{-1}$

c. $\mathbf{x} = \frac{\mathbf{b}}{A}$

d. $\mathbf{x} = A^{T}\mathbf{b}$

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12 If A is a 4×4 matrix with rank 3, how many solutions does $A\mathbf{x} = \mathbf{0}$ have?

a. Infinitely many

b. Exactly one

c. None

d. 3

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